

# Network Science

---

## Lecture 01: Introduction

Prof. Dr. Michael Granitzer

Dr. Kanishka Ghosh Dastidar

Dr. Jelena Mitrovic

# Welcome to Network Science

Modelling, analysing and inferring digital and real-world phenomena through networks.

## Today

- what is a network?
- why do networks matter?
- what can we analyse with networks?
- what will you learn in this course?

# Networks

**Guiding Question:** What do you think a network is?

# Your Everyday Networks

Think about networks you encounter every day.

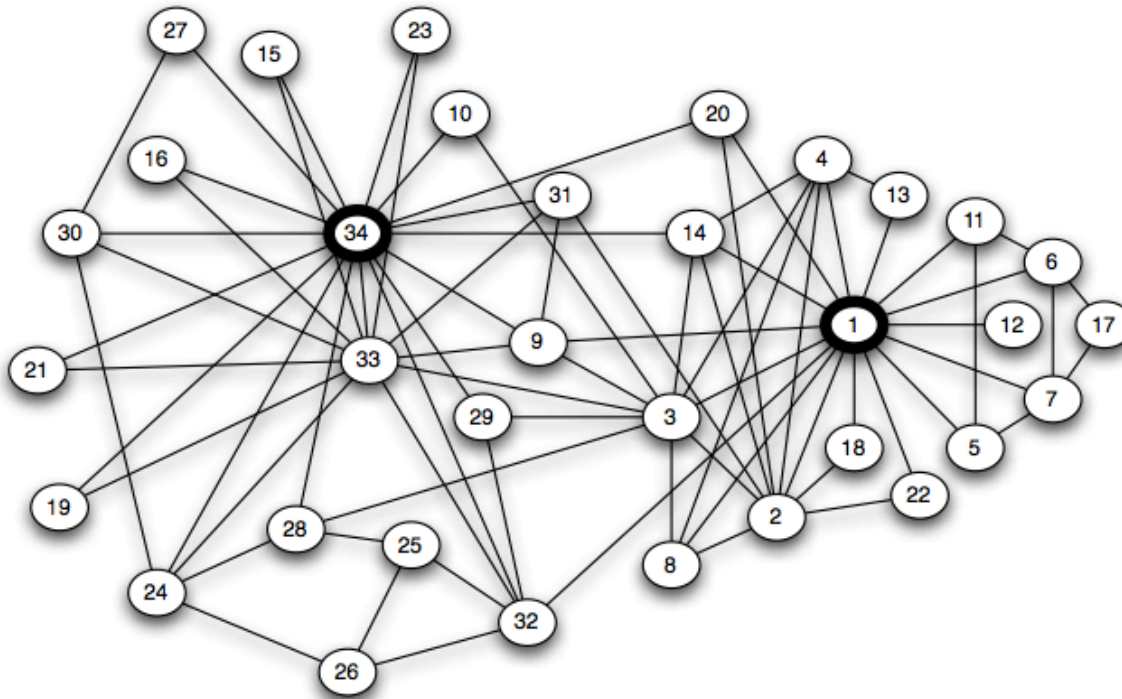
# Your Everyday Networks

Think about networks you encounter every day.

- Who did you talk to today? → **social network**
- Which websites link to each other? → **information network**
- Which companies trade with each other? → **economic network**
- Which roads connect which cities? → **infrastructure network**

# Social Networks

Link structures can reveal friendships, communities, and influence.



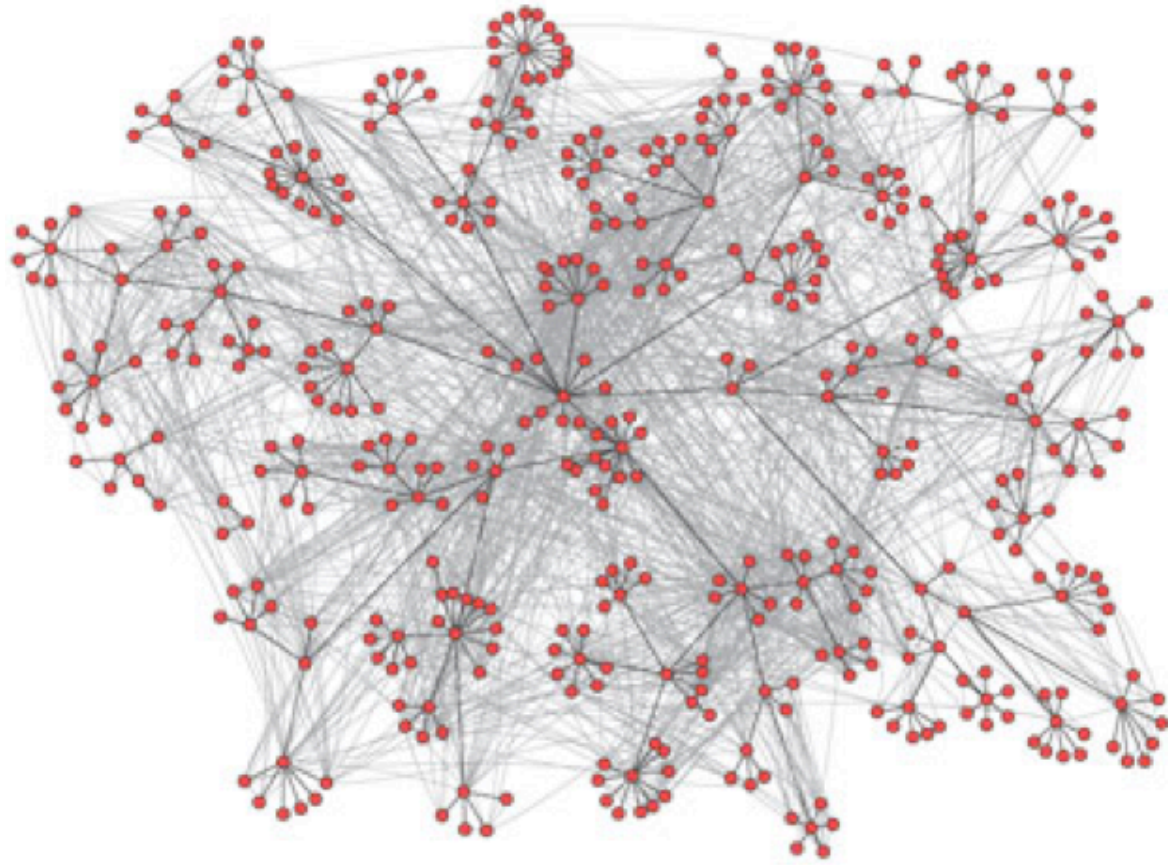
*Source: Zachary 1977 — modified; Original data: Zachary 1977; visualization redrawn*

## **Zachary's Karate Club (1977).**

Each node is a club member, each edge a friendship. The club split into two factions — visible as two dense clusters. One of the most-studied small social networks.

# Communication Networks

Link structures can reveal information flow and potential dissemination routes.



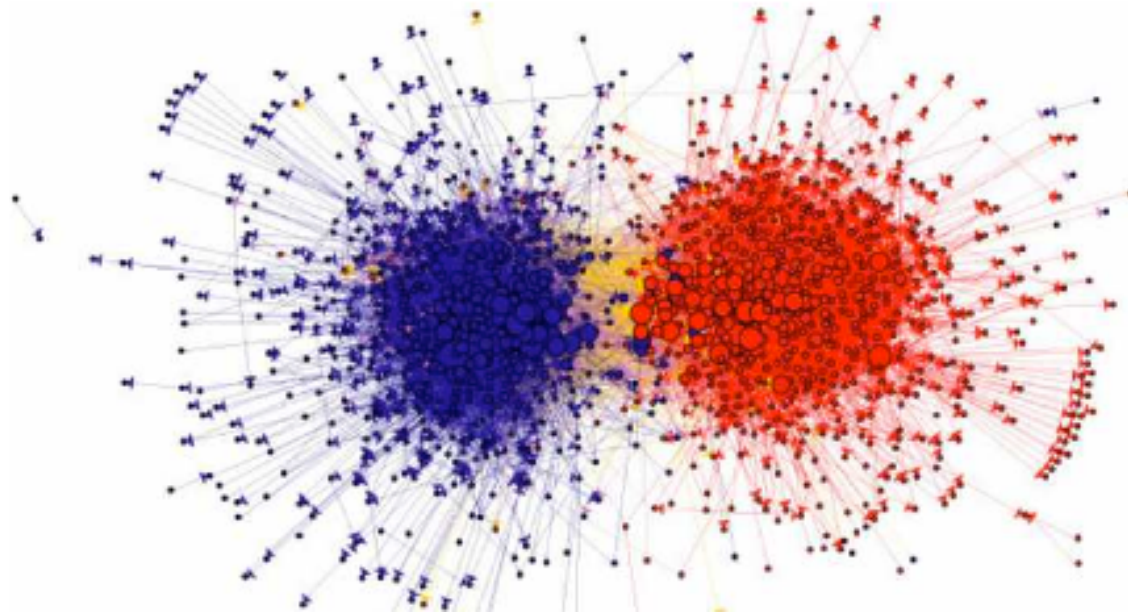
*Source: Easley & Kleinberg 2010, p. Ch. 2*

## **Email communication in an organization.**

Nodes represent employees, edges represent email exchanges. The actual communication topology differs substantially from the formal hierarchy.

# Information Networks

Link structures can reveal clusters, polarization, and community structure — before reading a single word.



*Source: Adamic & Glance 2005*

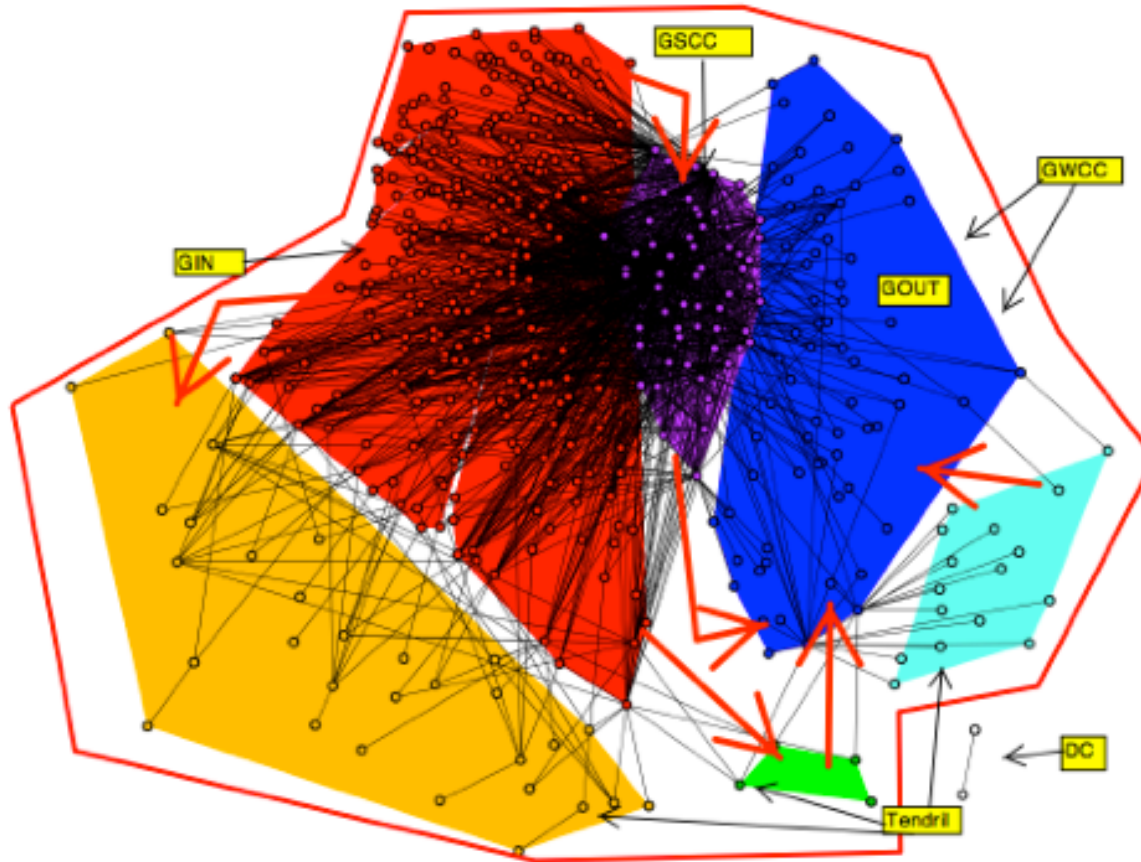
## Political blog network (2004 US election).

Nodes are political blogs, edges are hyperlinks between them. The two dense clusters correspond to liberal and conservative blogs — structure reveals political alignment without reading any content.



# Economic Networks

Financial systems form networks of dependencies and influence (Easley & Kleinberg, 2010).



Source: Easley & Kleinberg 2010, p. Ch. 1

## Interbank loan network.

Nodes are financial institutions, edges represent lending relationships. When one institution defaults, the impact cascades through the network — structure determines systemic risk.

# What Do All These Examples Have in Common?

# What Do All These Examples Have in Common?

- There are **things** (people, pages, institutions, cities).
- There are **connections** between them (friendships, links, loans, roads).
- The pattern of connections creates **structure** that matters.

# What Do All These Examples Have in Common?

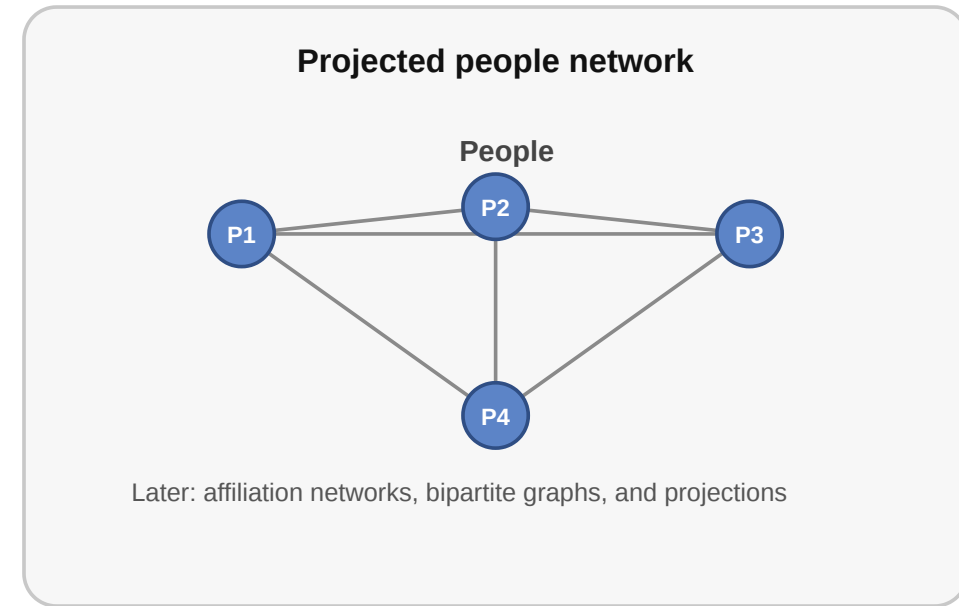
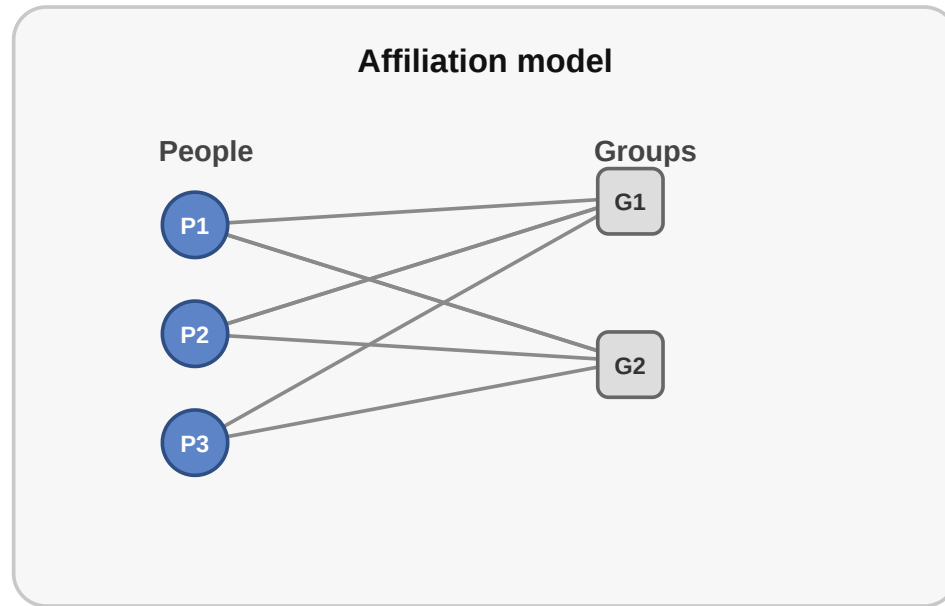
- There are **things** (people, pages, institutions, cities).
- There are **connections** between them (friendships, links, loans, roads).
- The pattern of connections creates **structure** that matters.

Network = Entities + Relationships between them

# What Gets Lost in a Network Model?

When we turn a real situation into a network, we always leave something out.

Same situation, different network models



The same relational setting can be modeled in more than one way. An affiliation structure can become a bipartite network or a projected one-mode graph depending on the question.

# Takeaway

A network is a structured collection of entities and the relationships between them. Understanding networks means understanding how to manage, describe, and reason about these entities and their connections.

# Quick Check

Which of the following statements about networks are correct?

- A network consists of entities (nodes) and relationships between them (edges)
- The same network abstraction applies across very different domains
- Networks always have directed edges
- A network model captures every detail of the underlying system
- Edges can carry additional information such as weight or type

# Quick Check — Solution

Which of the following statements about networks are correct?

- **Correct:** A network consists of entities (nodes) and relationships between them (edges)
- **Correct:** The same network abstraction applies across very different domains
- **Incorrect:** Networks always have directed edges (edges can also be undirected)
- **Incorrect:** A network model captures every detail of the underlying system (they are models and have lossy abstractions)
- **Correct:** Edges can carry additional information such as weight or type



# Impacts of Networks

---

**Guiding Question:** Why do networks matter — what can we learn from connections?

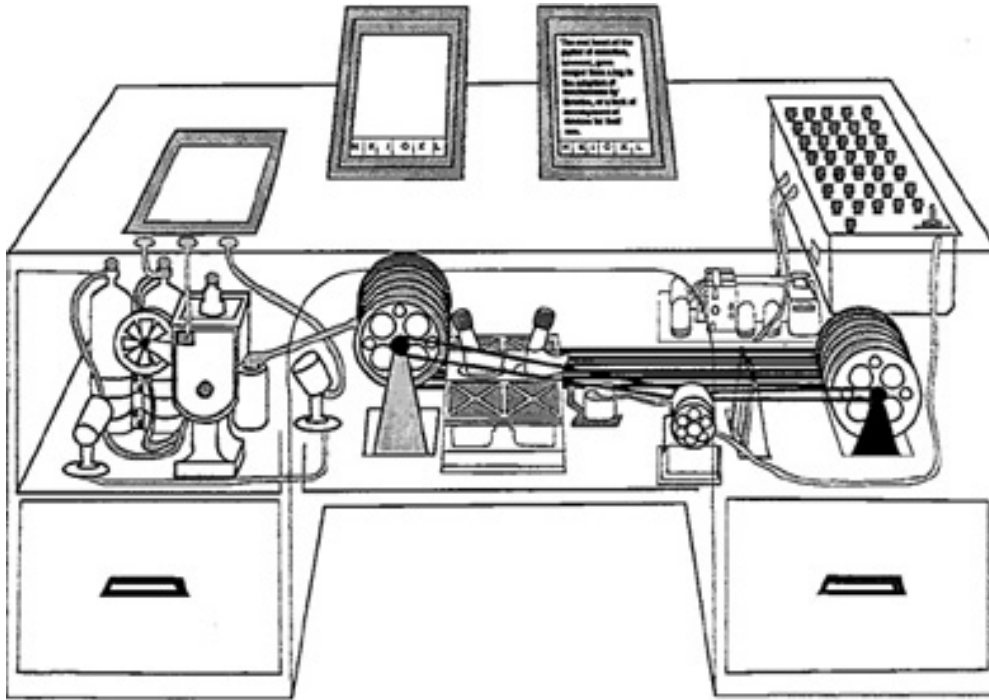
# The Whole Is More Than Its Parts

- Knowing every person in a city does not tell you how information travels — you need to know **who talks to whom**.
- Knowing every website does not tell you which ones are easy to find — you need to know **which pages link to which**.
- Knowing every bank does not tell you which failure would cause a cascade — you need to know **who has financial exposure to whom**.

Outcome  $\neq$  actor properties alone

# Practical Example: Linked Information

Vannevar Bush's MEMEX anticipated linked information systems and associative navigation (Bush, 1945).



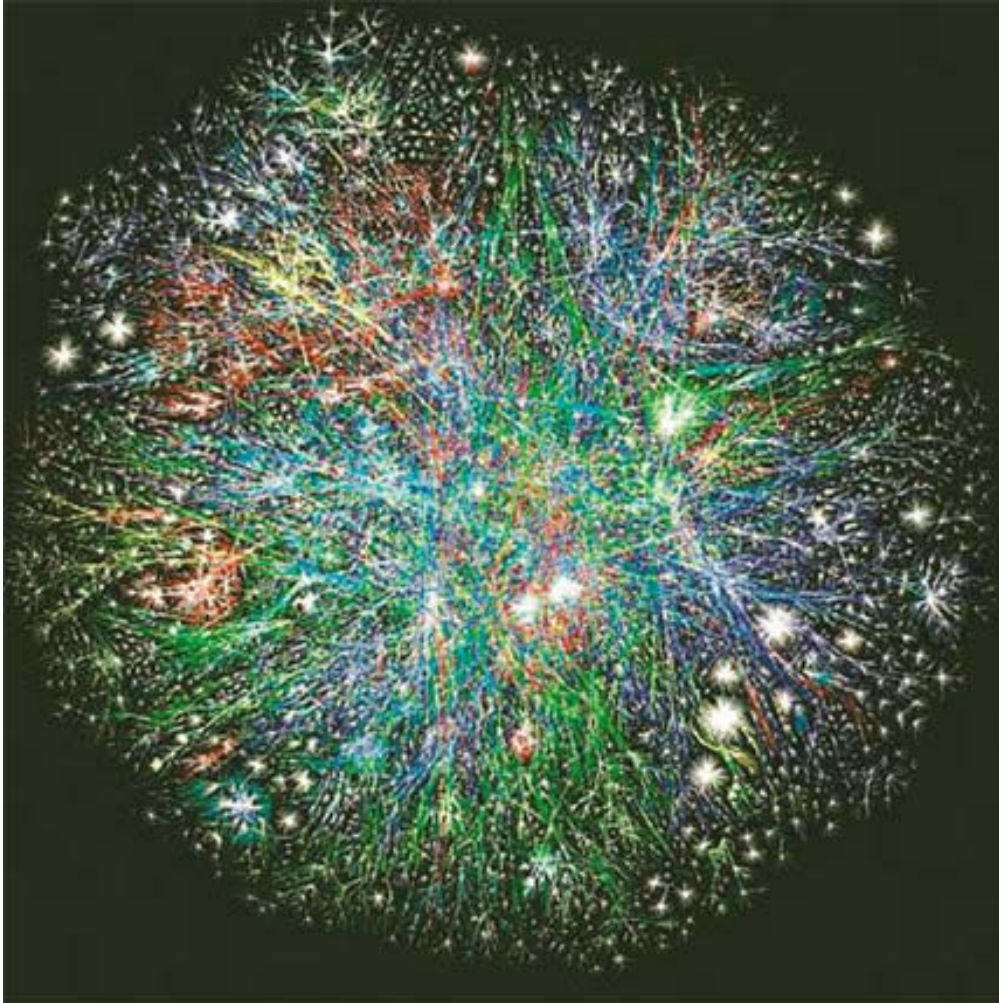
*Source: Bush 1945 — public-domain*

## **Vannevar Bush's MEMEX (1945).**

A conceptual device for storing and linking documents through associative trails. It anticipated the idea that information becomes valuable through connections — the intellectual ancestor of hypertext and the Web.

# The Web as a Network

The Web operationalized hypertext at planetary scale: linked documents, user navigation, and a growing information network (Berners-Lee, 1990).



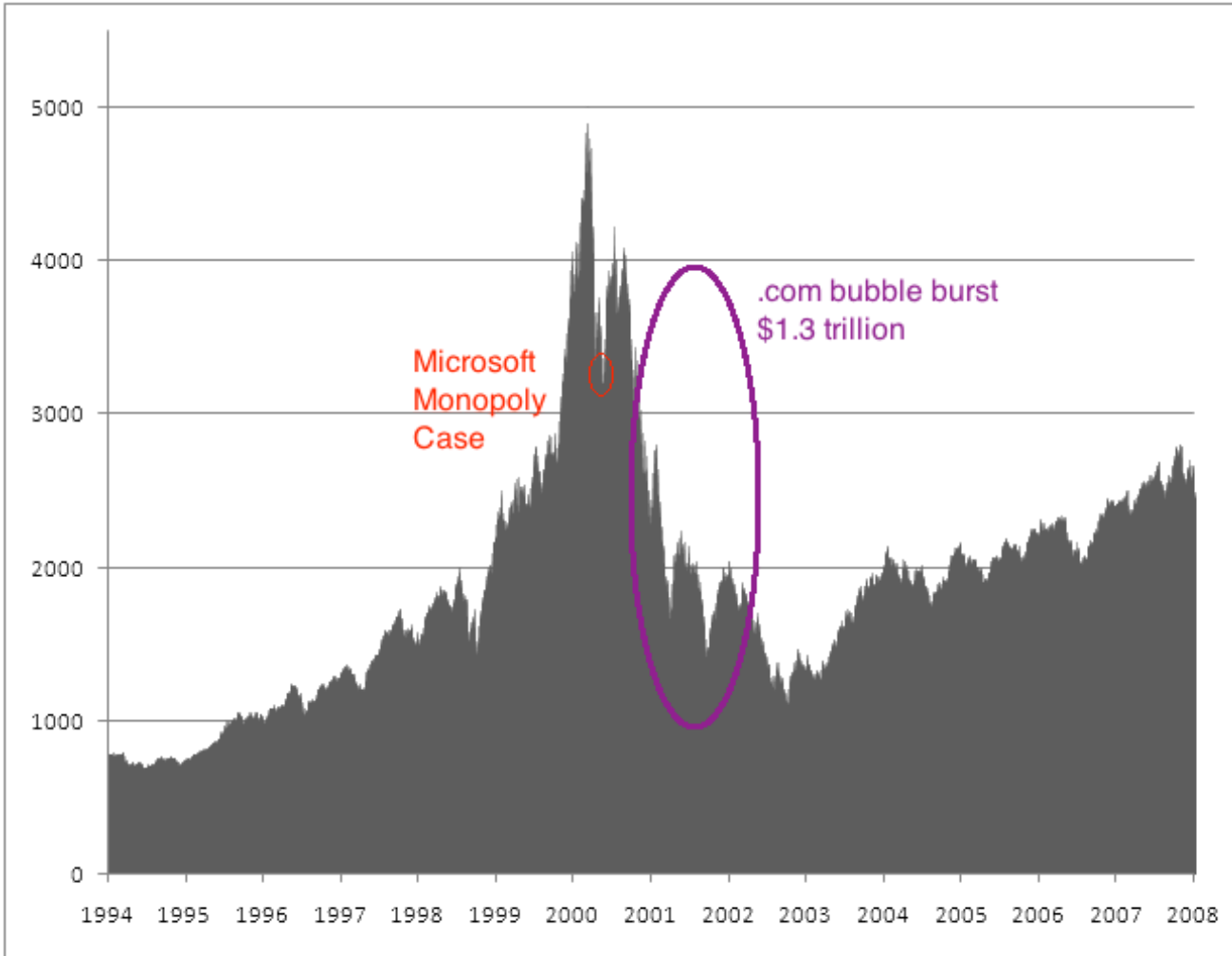
## Visualization of internet routing paths.

Each node represents a router or autonomous system, edges are data connections. The dense core and sparse periphery illustrate the uneven topology of real technical networks.

Source: Easley & Kleinberg 2010

# Practical Example: Platform Power

Surviving web companies learned to exploit network effects — users became part of the value-creation process (O'Reilly, 2005).



Source: Public market data, NASDAQ composite index

**NASDAQ composite index around the dot-com bubble (1995–2005).**

The sharp rise and crash illustrate how many early web companies failed. The survivors — Google, Amazon, Facebook — were those that understood and exploited network effects.

# What Can We Learn from Networks?

- Who is important? → centrality, influence
- Are there groups? → communities, clusters
- How do things spread? → diffusion, contagion
- How robust is the system? → fragility, resilience
- What shapes attention? → visibility, ranking, filtering

# Takeaway

Networks matter because structure shapes visibility, coordination, influence, and lock-in. Looking at connections reveals what looking at individuals alone cannot.

# Quick Check

Why are network models useful for studying platforms like the Web?

- A. They allow us to ignore structure and focus on individual users.
- B. They reveal how connections create effects like lock-in and influence.
- C. They primarily help in analyzing the content of individual web pages.
- D. They show that all users have equal influence in a system.



# Quick Check — Solution

Why are network models useful for studying platforms like the Web?

**Correct:** B. They reveal how connections create effects like lock-in and influence.

The Web's power comes not just from the individual pages (nodes), but from how they are linked (edges), which drives visibility and platform power.

# Network Science & Network Analysis

**Guiding Question:** What phenomena can we actually analyse with networks?

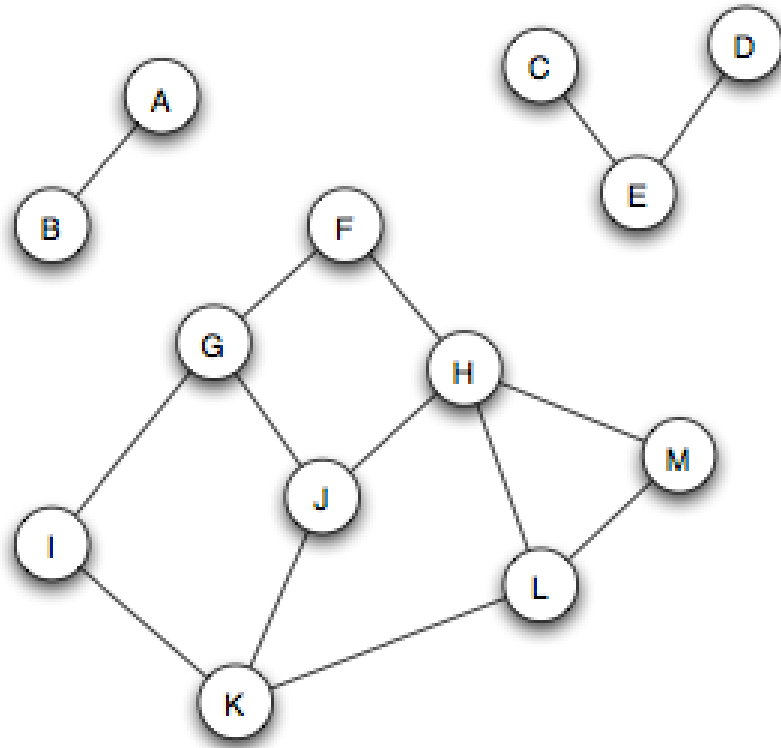
# From Intuition to Analysis

We already said networks reveal structure. But what kinds of questions can we actually answer?

- **Structure:** How is the network organized? Are there clusters, hubs, or bottlenecks?
- **Position:** Who sits in a central or peripheral position? Who bridges groups?
- **Dynamics:** How does information, disease, or opinion spread through the network?
- **Evolution:** How does the network change over time? What drives growth?

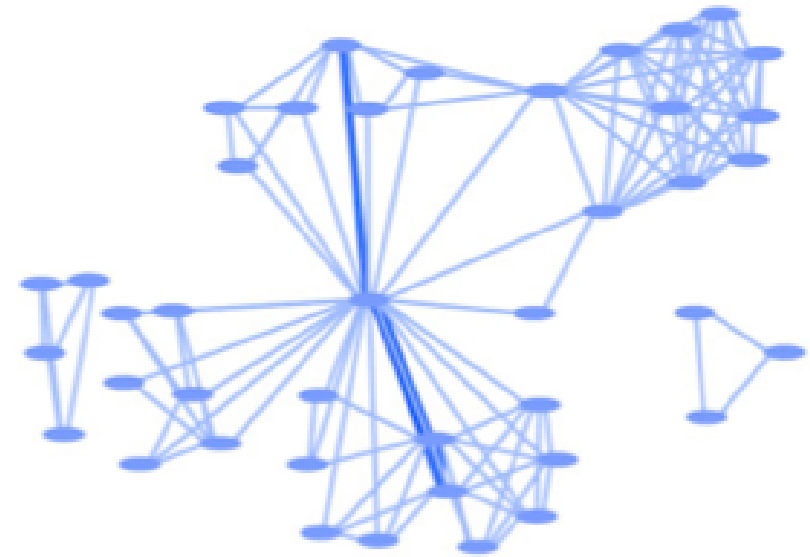
# Analysing Structure: Communities

Networks often have dense internal clusters that correspond to real groups.



*Source: Easley & Kleinberg 2010*

Three separate connected components — groups of nodes with no links between them.

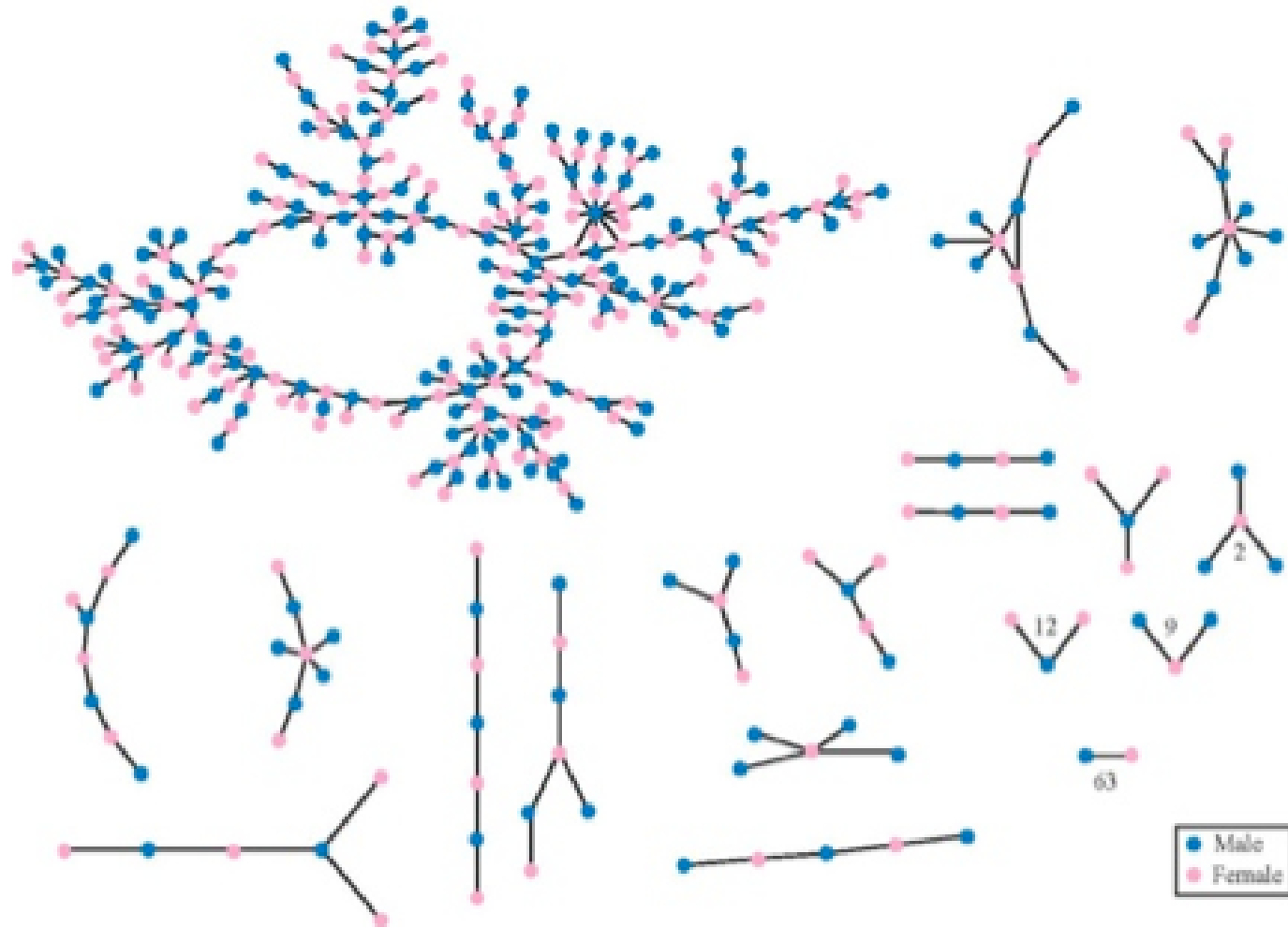


*Source: Easley & Kleinberg 2010*

A scientific collaboration network where clusters represent research groups.

# Analysing Dynamics: Diffusion and Spreading

How a rumor, virus, or innovation spreads depends critically on network structure.



Source: Easley & Kleinberg 2010

## High school friendship network.

Nodes are students, edges are reported friendships. The structure shows a dense core and peripheral chains — a rumor starting in the core spreads far faster than one starting at the edge.

# Core Questions of Network Science

Across all these domains, network science asks a common set of structural questions:

- How do we **describe** a network formally?
- Which structural **properties** matter?
- How do networks shape **diffusion** and coordination?
- Which common **patterns** appear across different domains?

# Takeaway

Network science gives us concrete tools to analyse structure, position, dynamics, and evolution. The same analytical questions apply whether we study friendships, web links, financial flows, or disease transmission.

# Quick Check

Match the network phenomenon with the type of analysis used to study it:

1. Finding groups of friends in a high school
2. Measuring how fast a rumor spreads
3. Identifying the most influential blogger

A. Dynamics (Diffusion) B. Structure (Communities) C. Position (Centrality)



# Quick Check — Solution

Match the network phenomenon with the type of analysis used to study it:

- 1 - B. Structure (Communities)
- 2 - A. Dynamics (Diffusion)
- 3 - C. Position (Centrality)

# Curriculum

**Guiding Question:** What do you want to learn about networks?

# What You Will Learn

## Course Topics

1. **Graph theory** — the formal language for describing networks
2. **Social networks** — ties, communities, and information flow
3. **Centrality and influence** — who matters and why
4. **Community detection** — finding groups in networks
5. **Information networks and the Web** — links, search, and ranking
6. **Network dynamics** — diffusion, contagion, and cascades

# Course Goal

Use network analysis to understand and design social information systems.

- not just learn definitions — **interpret** formal concepts
- not just run algorithms — **critique** the models behind them
- not just describe networks — **reason** about their consequences

# Recommended Reading

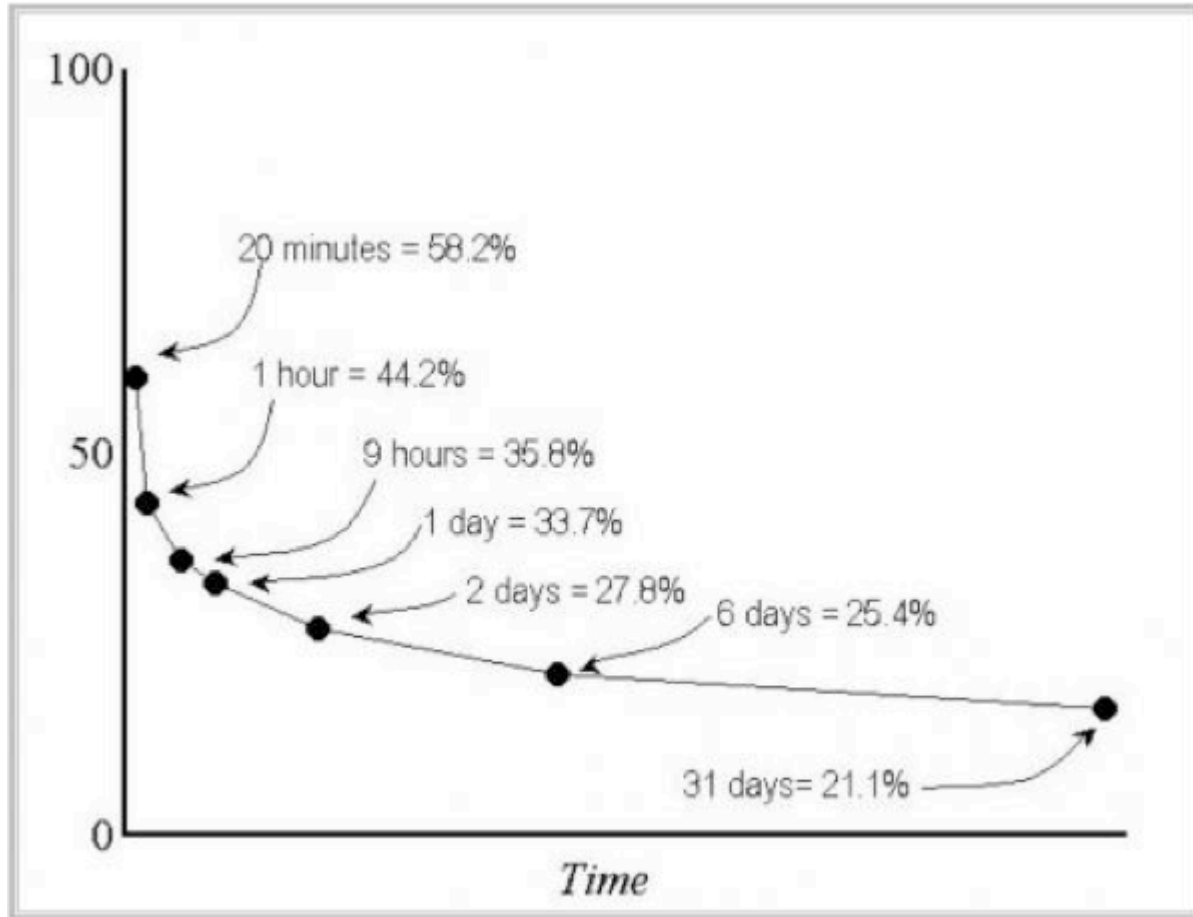
- David Easley and Jon Kleinberg, *Networks, Crowds, and Markets* (Easley & Kleinberg, 2010)
- Mark Newman, *Networks: An Introduction* (Newman, 2010)

# How to Learn This Course

- **preview** slides before class
- **discuss** key ideas in small groups
- **work through** exercises yourself before comparing solutions
- **use AI tools** to accelerate access to explanations — but if you cannot explain a concept without the tool, you do not yet own it

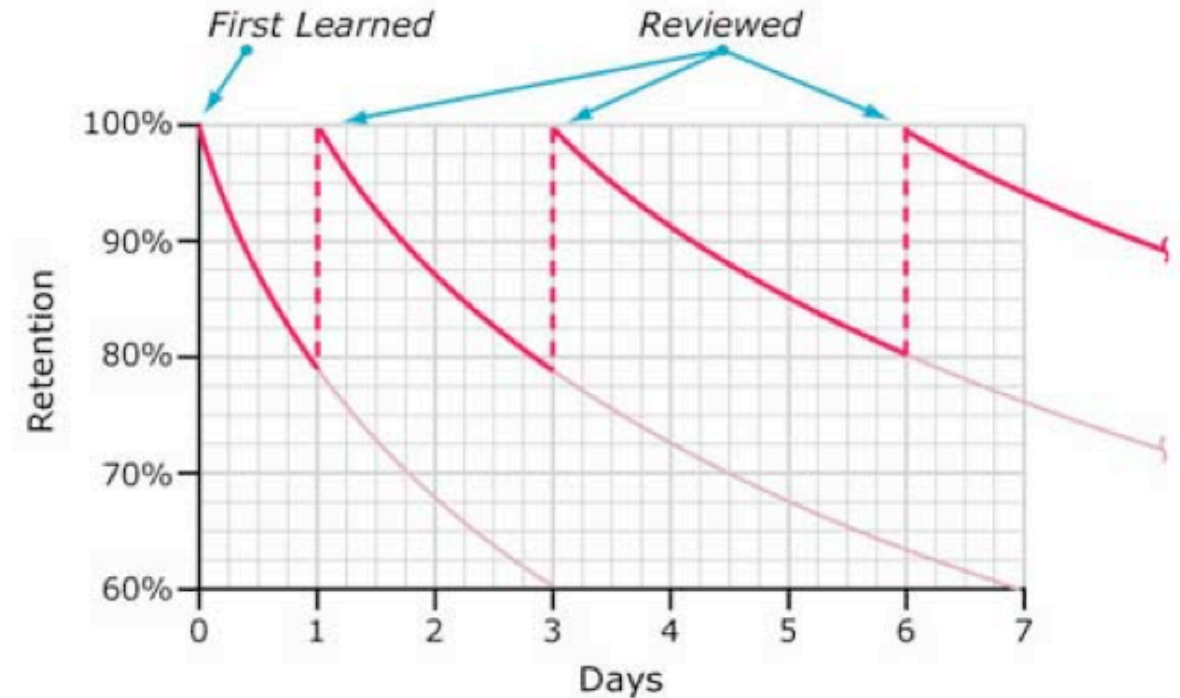
# Forgetting and Repetition

Repeated contact with the same concept reduces forgetting and improves retention.



Source: Ebbinghaus forgetting curve, redrawn

Typical Forgetting Curve for Newly Learned Information



Source: Ebbinghaus curve with spaced repetition, redrawn

# Takeaway

The course is about modeling, interpretation, and critique — not only about definitions. You will build a toolkit for understanding networks that applies across social, technical, and economic systems.





## Image Credits & References

Adamic, Lada A. and Glance, Natalie (2005). The political blogosphere and the 2004 U.S. election: divided they blog. *Proceedings of the 3rd International Workshop on Link Discovery (LinkKDD '05)*.

Bush, Vannevar (1945). As We May Think. *The Atlantic Monthly*.

Easley, David and Kleinberg, Jon (2010). Networks, Crowds, and Markets: Reasoning About a Highly Connected World. *Cambridge University Press*. <https://www.cs.cornell.edu/home/kleinber/net-works-book/>

Zachary, Wayne W. (1977). An Information Flow Model for Conflict and Fission in Small Groups. *Journal of Anthropological Research*.